

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1-Scenario Based

Course Code:	DSA0405	Course Title:	FUNDAMENTALS OF DATA SCIENCE
CO Assessed:	CO4-AT1-Scenario Based	Assessment:	Assessment Tool 1– Scenario Based
Weightage:	40% of CIA	Total Marks:	100
CO3 Statement:	Analyze and extract image features using detection and matching techniques for effective image representation and comparison. (BL4)		
Student Name:	Asish.s.s	Reg. No.:	192424348
Section / Batch:	BTech AI&DS	Date:	20-08-2026

Code of Conduct

I __Asish.s.s./192424348__ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly	
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts	
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning	

Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided	
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand	

INDUSTRY BASED PROBLEM-SOLVING

Confidence Interval and Hypothesis Testing Using Confidence Interval

1. Online Learning Platform — Confidence Interval

Problem-Solving Question:

An online learning platform measures the completion time of a course for 100 students. The mean completion time is 14 days and the standard deviation is 3 days. Construct a 95% confidence interval for the true mean completion time.

Solution:

Given: $n = 100$, sample mean $\bar{x} = 14$ days, sample standard deviation $s = 3$ days.

Since the population standard deviation is not given, use the t-distribution with $df = 99$. For a 95% confidence level, $t_{0.025,99} \approx 1.984$.

Standard Error (SE) = $s/\sqrt{n} = 3/\sqrt{100} = 0.30$ days.

Margin of Error = $t \times SE = 1.984 \times 0.30 = 0.5952$ days.

95% Confidence Interval = $\bar{x} \pm \text{Margin of Error} = 14 \pm 0.5952$.

95% CI = (13.4048 days, 14.5952 days) $\approx$ (13.40 days, 14.60 days).

Interpretation: We are 95% confident that the true mean course completion time is between approximately 13.40 and 14.60 days.

2. Hospital — Confidence Interval

Problem-Solving Question:

A hospital surveys 400 randomly selected patients, and 340 report that they are satisfied with the service. Construct a 95% confidence interval for the proportion of satisfied patients.

Solution:

Given: $x = 340$ satisfied patients and $n = 400$.

Sample proportion $\hat{p} = x/n = 340/400 = 0.85 = 85\%$.

Standard Error = $\sqrt{[\hat{p}(1-\hat{p})/n]} = \sqrt{[(0.85)(0.15)/400]} \approx 0.01785$.

Using $z = 1.96$ for a 95% confidence level:

Margin of Error = $1.96 \times 0.01785 \approx 0.03499$.

95% Confidence Interval = 0.85 ± 0.03499 .

95% CI = (0.8150, 0.8850) $\approx$ (81.50%, 88.50%).

Interpretation: We are 95% confident that the true proportion of patients satisfied with the hospital service is between approximately 81.50% and 88.50%.

3. Telecom — Confidence Interval

Problem-Solving Question:

A telecom company wants to estimate the average monthly data usage of its customers. A sample of 225 customers has a mean usage of 16 GB and a standard deviation of 5 GB. Construct a 99% confidence interval and interpret the result.

Solution:

Given: $n = 225$, sample mean $\bar{x} = 16$ GB, sample standard deviation $s = 5$ GB.

Since the population standard deviation is not given, use the t-distribution with $df = 224$. For a 99% confidence level, $t_{0.005, 224} \approx 2.601$.

Standard Error = $s/\sqrt{n} = 5/\sqrt{225} = 5/15 =$ **0.3333 GB**.

Margin of Error = $2.601 \times 0.3333 =$ **0.8670 GB**.

99% Confidence Interval = 16 ± 0.8670 .

99% CI = (15.1330 GB, 16.8670 GB) $\approx$ (15.13 GB, 16.87 GB).

Interpretation: We are 99% confident that the true average monthly data usage of the company's customers lies between approximately 15.13 GB and 16.87 GB.

4. Customer Support — Hypothesis Testing Using CI

Problem-Solving Question:

A company claims that the average time taken by its customer-support team to resolve a complaint is 24 hours. A sample produces a 95% confidence interval of (25.2, 27.8) hours. Based only on this confidence interval, would you reject the company's claim? Explain.

Solution:

Claimed population mean: $\mu = 24$ hours.

Given 95% confidence interval: (25.2, 27.8) hours.

The claimed value of 24 hours is **not contained** within the 95% confidence interval.

Therefore, based only on this confidence interval, we would **reject** the company's claim at the 5% significance level.

Interpretation: The sample evidence suggests that the true average complaint-resolution time is greater than 24 hours. Since the entire confidence interval is above 24 hours, the claimed mean of 24 hours is not consistent with the sample evidence at the 95% confidence level.

5. Manufacturing — Hypothesis Testing Using CI

Problem-Solving Question:

A manufacturer claims that the average weight of a packaged product is 500 grams. A 95% confidence interval based on a sample is (496, 502) grams. Is the claimed value of 500 grams consistent with the sample evidence? Explain your answer.

Solution:

Claimed population mean: $\mu = 500$ grams.

Given 95% confidence interval: (496, 502) grams.

The claimed value of 500 grams **lies inside** the 95% confidence interval.

Therefore, 500 grams is a plausible population mean based on the sample evidence.

Conclusion: The manufacturer's claim is consistent with the sample evidence at the 95% confidence level. We would not reject the claim based on this confidence interval alone.